Viva and Interview Questions

What is the project? A document-grounded AUTOSAR HLD assistant for engineering Q&A and structured SWC/Port/Interface exploration.

Why RAG? Index changing PDFs and retrieve source evidence without retraining model weights; expose source citations.

Which AI algorithms? Pretrained transformer embeddings (all-MiniLM-L6-v2), vector similarity retrieval, and a Gemini generative model.

Did you train a model or apply PCA? No. The project uses pretrained models; no PCA/t-SNE/UMAP stage is implemented.

What are preprocessing steps? PyMuPDF block/page extraction, header/footer and hyphen cleaning, whitespace normalization, hierarchical chunking.

What is top-k? Number of candidate nearest chunks (5 by default).

What is the relevance threshold? Minimum retrieval similarity if enabled. Application default is None, so not always applied.

Why two databases? ChromaDB for semantic chunks/embeddings; SQLite for structured SWCs/Ports/Interfaces.

What exactly does Gemini do? Converts relevant retrieved HLD text into a readable answer; it is not the PDF retriever.

What does citation verification prove? Only that cited IDs point to retrieved source; a separate human check is needed for actual claim support.

What is an extra feature over simple PDF chatbot? Domain-aware entity extraction and architecture explorer, source provenance, deterministic citations and offline MockLLM testing.

How do you evaluate? Tests, golden questions, retrieval/citation-ID metrics, human factual support, refusal correctness and latency.

What failed during testing? A generic MockLLM answer with a valid citation was observed even for a missing-MCU question; it is a limitation, not a successful negative-grounding demonstration.

What next? Hybrid lexical+vector search for AUTOSAR identifiers, ARXML support, graph relations and stronger factual-entailment checks.